

Review of WBER-2025-387 “Stress-Testing Survey-to-Survey Imputation: Understanding When Poverty Predictions can Fail”

This is a useful exposition of survey-to-survey (S2S) imputation, with simulations based on an assumed data-generating process demonstrating when S2S is likely to succeed or when its predictions could be misleading. Given the many ways in which predictions can go wrong I wonder if researchers would recommend using the method, given that any prediction could easily be challenged using one or more negative examples in the article.

Examples fall in two categories: contemporary imputations (sections 4.2 and 4.4), and imputation over time (section 4.5), where structural changes over time only add to problems that could arise with contemporary imputation.

I find the use case for contemporary imputations (i.e., when source and target surveys cover the same population at approximately the same time) questionable. The authors mention one potential but unconvincing application on p11. (See below) On the other hand, separating contemporary and intertemporal examples very much helps the presentation in the article.

Given the many problems that could arise in S2S imputation communicating results to end users will be very difficult. The Abstract promises guidance on ‘how its limitations should be communicated’ but I don’t think the authors deliver on this point, despite a long list of recommendations in the concluding Section 5.

Below I give a list of specific comments, many of editorial nature, without expecting that authors address all comments. Page numbers refer to the numbers as provided by the WBER at the top of the page, i.e. bottom-of-the-page-number plus 1.

Specific comments

- p3: Perhaps mention explicitly the choice for a monetary poverty concept. This may not be an obvious choice for all readers
- p6,l20: what are granular predictors?
- p7, l40: ‘and noted by the authors’????
- p7, l52: Presents S2S imputation as a tool for predicting poverty for the entire population. This conflicts with p11, l7-11, where an application is suggested where stunting rates are estimated across welfare quintiles
- p8, l9: confusing exposition; e_i will generally not be independent of x_i , so β is not the coefficient vector of interest, but β plus the linear projection of e on x (which is the vector estimated by OLS)
- p8, l33: ‘welfare vectors’. Explain. I don’t think this is clear to all readers
- p8, l44-46 K is the number of covariates, including the constant term
- p11, l9-11: This is the single example given where contemporary S2S imputation could be useful, by estimating the joint distribution of welfare and stunting: ‘... would allow for accurate estimation of stunting rates across welfare quantiles...’. If stunting is absent in the source survey and welfare is imputed into the target (DHS) survey (which does include stunting information), a cross table can be compiled but it would be unreliable. A sufficient condition for identification would be that, given covariates, stunting and welfare are independent, which is rather unlikely. For situations like this in the context of multiple imputation, see e.g. Susanne Rässler (2004).
- p11, l22: ‘depends critically’: the three assumptions mentioned can apparently be relaxed, so they aren’t critical after all?
- p11, l31: Errors will be zero-centered by construction (being OLS residuals), but why do they need to be symmetric around zero?
- p12, l6: ‘rigorously test’: ok, but how?
- p12, l26 ‘same as β ’????
- p12, l25-38: isn’t this too obvious to merit an 11 line paragraph?
- p12, fn 15: so what? the effects of $X'X$ and $X'\beta$ will cancel if β is constant
- p12, l56: independence between $\bar{X}\beta$ and e : $\bar{X}\beta$ is a constant and therefore independent of any random variable

- p13, l6: this formula for poverty requires $x_i\beta$ to be normally distributed as well
- p13, fn16: note that temporal changes in the correlation between error term and (observed) covariates (after OLS zero by construction in the source survey) are not conceptually different from changes in β and σ^2 . They are all structural changes
- p14, l1-8; that the estimated Gini can change with observed characteristics is a good thing, right? (I miss the point of this sentence.)
- p14, section 4.1; Why this particular DGP? There's only dependence between x_2 and x_3 (correlation 0.67) and the corresponding $x\beta$ is very close to normally distributed. This looks rather arbitrary and prompts the question of how generalizable the simulation results are
- p14, l47-48: why does the max of 1 and a Poisson random variable need to be rounded?
- p14, l56: 2^2 presumably being the variance and not the standard deviation?
- p15, l27: 'noise is estimated' : too colloquial to my taste; unnecessary jargon
- p15, l29-36; I don't fully understand this simulation set-up. Please clarify. p15, l32 mentions 'same data set'. If all imputation is within-sample then there is a clear risk of overfitting and results that are better than expected under the DGP
- p15, l57-59: I guess the MSE mentioned is the MSE of predicted $\ln y$, but the interest is in a probability, a point on the CDF of $\ln y$, which is quite different. I wouldn't expect minimizing one MSE would also minimize the other. So it is not clear to me what the authors want to say here.
- p16, l3: rank correlation: specify between which variables (y and $x\beta$ vs y and $x\beta + \varepsilon^*$)
- p16, l43: 'same variance' -> 'constant variance'
- p16, l47: I find the statement that 'all methods effectively replicate the underlying welfare distribution' too strong. It suggests much more than the statement that the methods have very small bias (if any). From p15, l35-36 I understand that Figure 2 is based on the average of applying the methods over 1000 experiments (i.e., populations). The Figure would be even more favorable if the number of populations would be further increased. But 'all methods effectively replicate the underlying welfare distribution' might be misread to mean that better than 0.03 percentage points precision can be achieved in practice, i.e. by a single experiment. To avoid misunderstanding it would be good to very clearly specify what is meant by unbiased methods in the context of S2S mapping.
- p17, around Fig 3: A conclusion might be added that drawing from empirical residuals is always a good idea (not only in the context of heteroskedastic errors see p19, l33)
- p17, l48-52: To include an experiment with normal errors in a section headed 'Under non-normally distributed errors' is a bit confusing
- Section 4.2.5: I guess that the irrelevance of nesting of errors is due to the fact that the results are the average of 1000 experiments. Yes, the methods are virtually unbiased, but results from a single experiment are definitely more imprecise when residuals are correlated than when they are independent. Please check again and adjust text if necessary
- Section 4.2.6: I don't see why the two changes of the DGP would highlight limitations of PMM in particular. Presumably, changing only the covariate distribution and leaving coefficients and error distribution unchanged (as in the first variation) would not affect the success of all the methods considered in the paper. Conversely, changing coefficients would negatively affect predictive power of all the methods. Please clarify why PMM is special in this regard
- Section 4.3, p24, l41-46 (and Appendix 6.2). Unbiasedness means that the expected deviation (or when simulating, the mean over many trials) is zero. But the expectation of the *squared* deviation is not. Also, the argument that the population is larger than the target sample is less convincing for two-stage surveys when precision is determined (mainly) by the number of primary sampling units, and only marginally by total population size.
- Section 4.4.3: Standardizing XB is very much like discarding the target survey, considering p13, equation (5) or the Gini equation on p12. I guess a similar comment can be made about matching on XB: it amounts to resetting the distribution of XB to that of the source survey.
- Section 4.5.: Imputation over time is in my opinion the real use case for S2S mapping (as discussed by the authors in chapter 2) but there is not much one can do about changes in structural parameters that cannot be estimated or even allow for plausible guesses. In this respect the conclusion on top of page 30 is more mantra than helpful guidance
- p30, l24: the authors might note that the no-bias point at percentile 50 arises from the symmetry of

the error distribution. See also Figure 1 on p16.

- p31, l3-4: Sloppy phrasing. An $x\%$ increase of the constant term additively increases *transformed* welfare by $x\%$ of the constant, and *untransformed* welfare grows (multiplicably) by $x\%$ times the constant term. If ‘transformed welfare for everyone’ increases by 1% all of $x\beta$ is multiplied by 1.01, not only the constant term.
- Section 5: The list of recommendations is useful but sobering. Are policy makers willing to consider a big pile of caveats? (Cf. Harry Truman asking for a one-armed economist.) S2S predictions will be easy to communicate but hard to explain.

Reference

Susanne Rässler. “Data Fusion: Identification Problems, Validity, and Multiple Imputation”, *Austrian Journal of Statistics*, Volume 33 (2004), p153-171.